Rosen — §2.6: Matrices

Selected Exercises

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Exercise 13

In this exercise we show that matrix multiplication is associative. Suppose that A is an $m \times p$ matrix, B is a $p \times k$ matrix, and C is a $k \times n$ matrix. Show that $A(BC) = (AB)C$.

Exercise 17

Let A and B be two $n \times n$ matrices. Show that

(a) $(A + B)^t = A^t + B^t$.

(b) $(AB)^t = B^t A^t$.

Exercise 19

Let A be the 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Show that if $ad - bc \neq 0$, then

$$A^{-1} = \begin{bmatrix} \dfrac{d}{ad - bc} & \dfrac{-b}{ad - bc} \\ \dfrac{-c}{ad - bc} & \dfrac{a}{ad - bc} \end{bmatrix}.$$

Exercise 21

Let A be an invertible matrix. Show that $(A^n)^{-1} = (A^{-1})^n$ whenever n is a positive integer.

Exercise 23

Suppose that A is an $n \times n$ matrix where n is a positive integer. Show that $A + A^t$ is symmetric.

Exercise 31

In this exercise we show that the meet and join operations are commutative. Let A and B be $m \times n$ zero–one matrices. Show that

(a) $A \vee B = B \vee A$.

(b) $B \wedge A = A \wedge B$.

Exercise 33

We will establish distributive laws of the meet over the join operation in this exercise. Let A, B, and C be $m \times n$ zero–one matrices. Show that

(a) $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$.

(b) $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$.

Exercise 35

In this exercise we will show that the Boolean product of zero–one matrices is associative. Assume that A is an $m \times p$ zero–one matrix, B is a $p \times k$ zero–one matrix, and C is a $k \times n$ zero–one matrix. Show that $A \odot (B \odot C) = (A \odot B) \odot C$.